Fake News Detection
Using Semantic Classification

Semantic Classification Assignment Report

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1. Introduction & Business Objective

The spread of fake news has become a significant challenge in today's digital world. With the massive volume of news articles published daily, it is increasingly difficult to distinguish between credible and misleading information. This creates a pressing need for automated systems that can classify news articles as true or fake, helping to reduce misinformation and protect public trust.

The objective of this assignment is to develop a Semantic Classification model using the Word2Vec method to extract semantic relations from news text and train supervised machine learning models to categorise articles based on their meaning. Three models were built and evaluated: Logistic Regression, Decision Tree, and Random Forest.

2. Data Description

Two datasets were used for this assignment:

Dataset	Description	Records
True.csv	True / legitimate news articles	21,417
Fake.csv	Fake / misleading news articles	23,502

Both datasets contain three columns: title (headline), text (article body), and date (publication date). The combined dataset has approximately 44,919 articles.

3. Data Preparation

The following steps were performed during data preparation:

Labelling: A new column "news_label" was added to both DataFrames. True news was labelled as 1 and fake news as 0.

Merging: Both DataFrames were concatenated into a single DataFrame using pd.concat() with ignore_index=True.

Null Value Handling: Null values were checked using isnull().sum(). Rows containing null values were dropped using dropna().

Column Merging: The "title" and "text" columns were combined into a new column "news_text" by joining them with a space. The original "title", "text", and "date" columns were then dropped.

4. Text Preprocessing

4.1 Text Cleaning

A clean_text() function was created to perform the following operations on each article:

- Convert all text to lowercase
- Remove text enclosed in square brackets (e.g., [Reuters])
- Remove all punctuation marks
- Remove words containing digits (e.g., dates, numbers)

4.2 POS Tagging & Lemmatization

Using spaCy's en_core_web_sm model, each cleaned article was processed through a POS tagging and lemmatization pipeline. The function:

- Tokenises the text and assigns Part-of-Speech tags to each token
- Filters out stopwords
- Retains only tokens tagged as NN (singular noun) or NNS (plural noun)
- Lemmatizes the retained tokens to their base form
- Joins the lemmatised tokens back into a single string

This step significantly reduces the text to its core semantic content -- the nouns that carry the most meaning for distinguishing between true and fake news.

5. Train-Validation Split

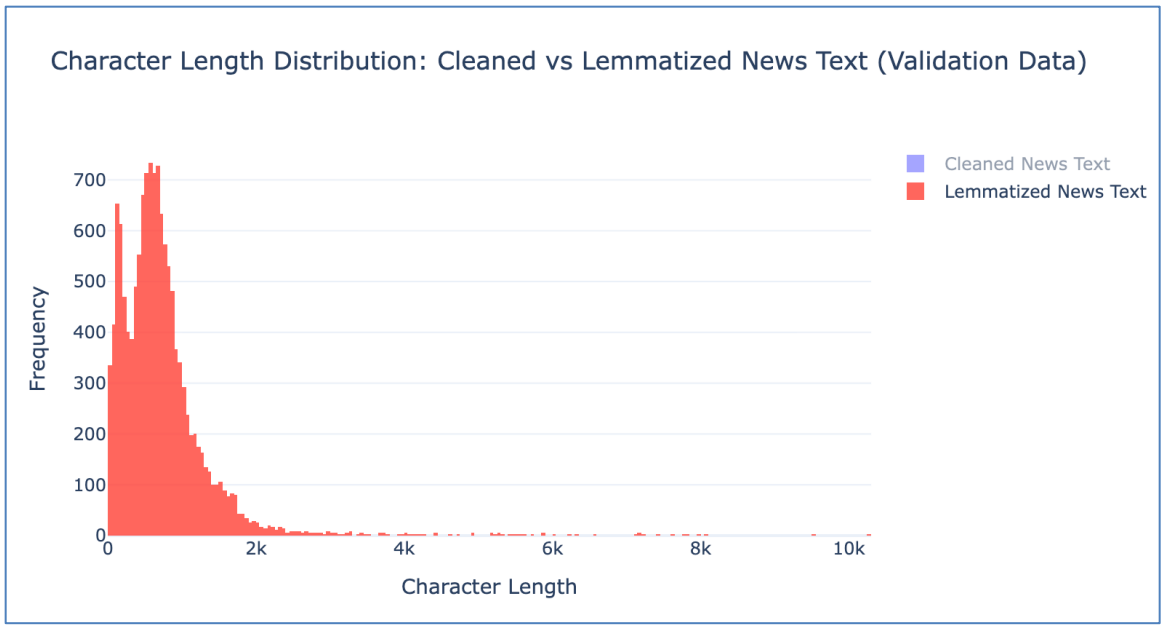
The cleaned and preprocessed DataFrame was split into 70% training data and 30% validation data using sklearn's train_test_split() with stratification on the "news_label" column to maintain the class distribution in both sets. A random_state of 42 was used for reproducibility.

6. Exploratory Data Analysis

6.1 Character Length Distribution

Histograms were plotted to compare the character lengths of the cleaned news text versus the lemmatized text (after POS tag filtering). Key observations:

The cleaned text has a broader distribution with longer character lengths, while the lemmatized text is significantly shorter. This confirms that the POS filtering step removes a substantial portion of non-noun content (typically 50-70% of the text).

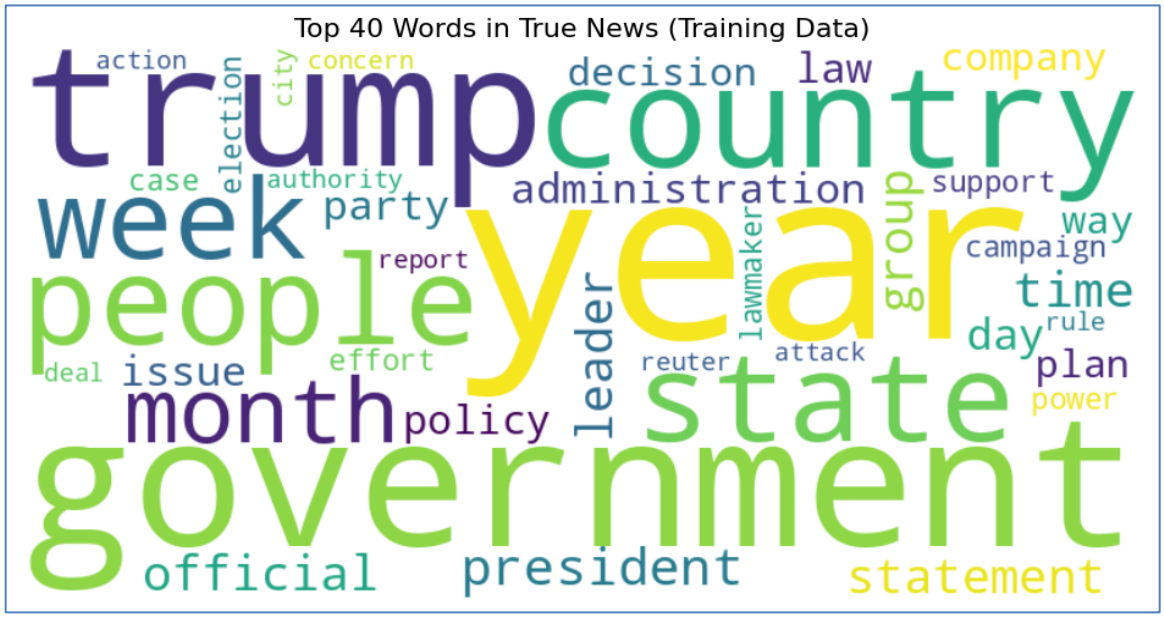


6.2 Word Clouds

Word clouds were generated to visualise the top 40 most frequent words in true and fake news articles from the training data.

Word Cloud - True News

The true news word cloud is dominated by formal, institutional terms such as "president", "state", "government", "official", "country", "law", "security", "party", "election", and "court". These reflect the structured, fact-based reporting style typical of legitimate news sources like Reuters and AP.



Word Cloud - Fake News

The fake news word cloud features more personality-driven and sensational terms such as "trump", "people", "media", "america", "woman", "video", "obama", "clinton". These words indicate a more opinion-driven, emotionally charged writing style with frequent references to specific personalities and social media content.



6.3 N-gram Analysis

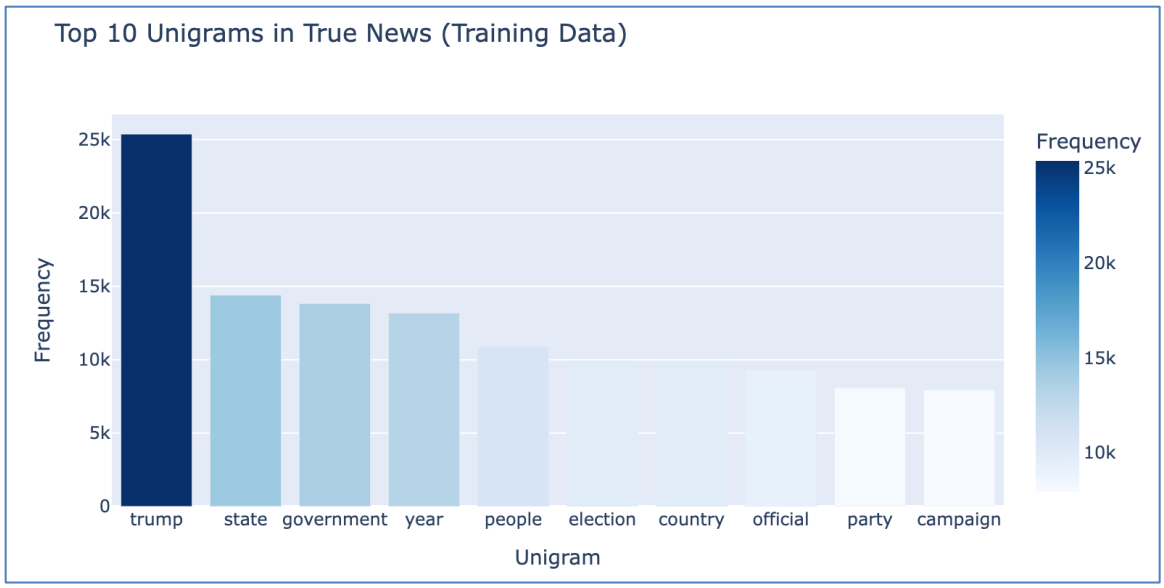
The top 10 unigrams, bigrams, and trigrams were extracted and plotted as bar charts for both true and fake news in the training data.

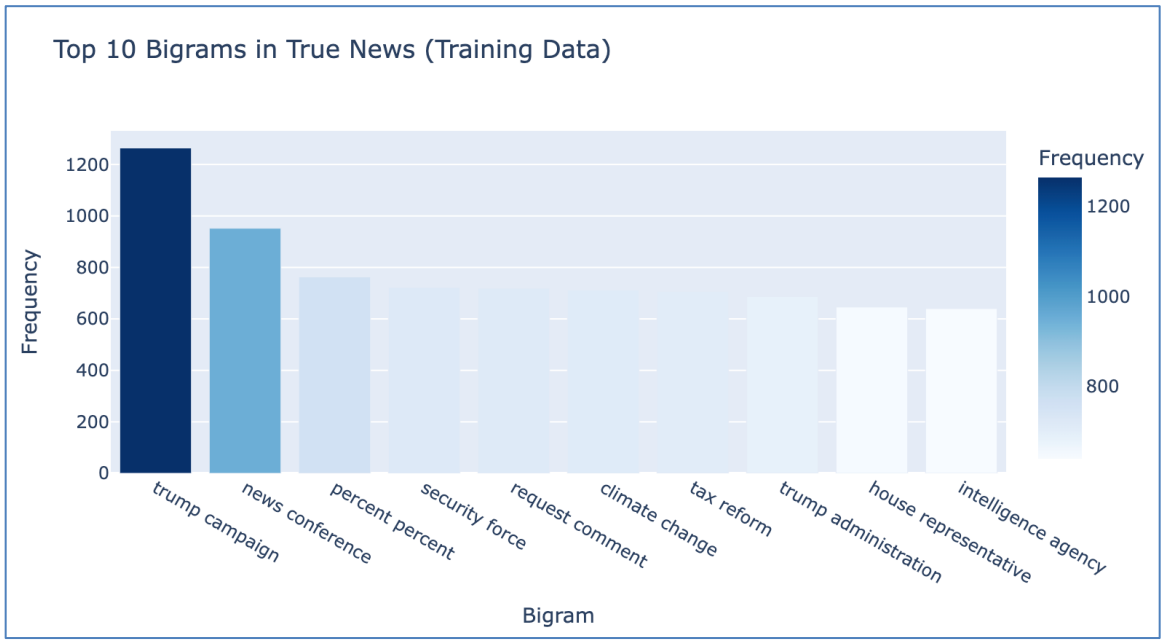
True News N-grams

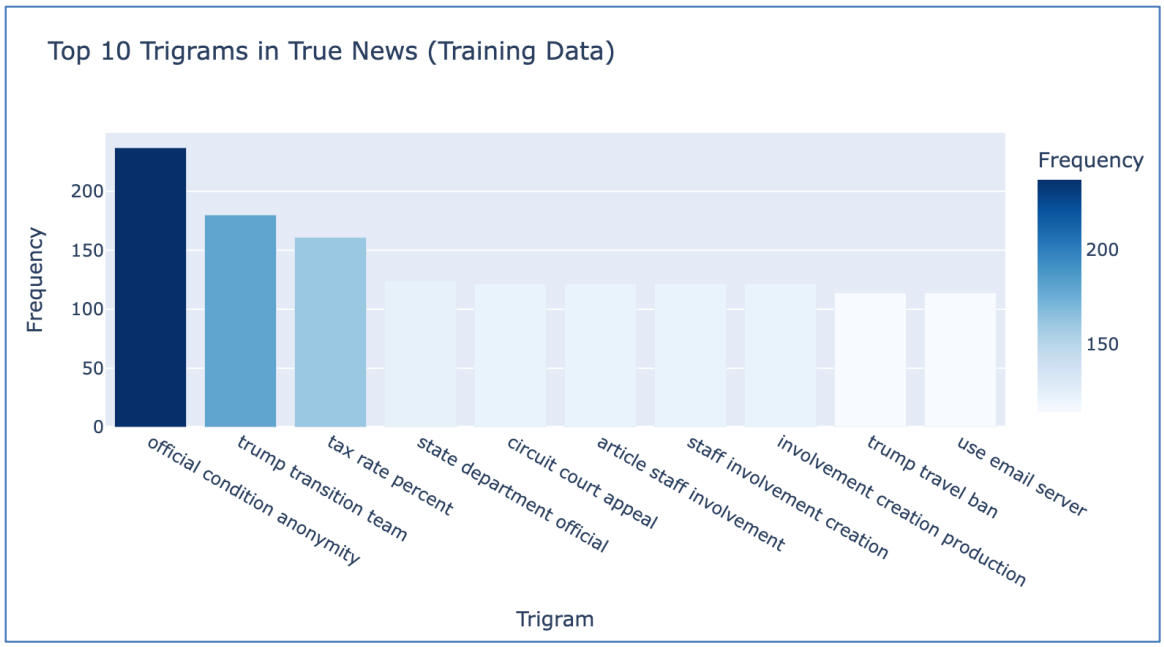
Unigrams: The most frequent unigrams in true news include "state", "president", "government", "official", "country", "party", reflecting political and governmental reporting.

Bigrams: Common bigrams include institutional phrases like "white house", "united state", "north korea", "supreme court", indicating coverage of major political institutions and events.

Trigrams: Frequent trigrams include "united state president", "north korea nuclear", and similar formal phrases characteristic of wire-service reporting.


Top 10 Unigrams in True News (Training Data)
Frequency
25k
20k
15k
10k
5k
0
trump
state
government
year
people
election
country
official
party
campaign
Unigram
Frequency
25k
20k
15k
10k


Top 10 Bigrams in True News (Training Data)
Frequency
1200
1000
800
600
400
200
0
trump campaign
news conference
percent percent
security force
request comment
climate change
tax reform
trump administration
house representative
intelligence agency
Bigram
Frequency
1200
1000
800

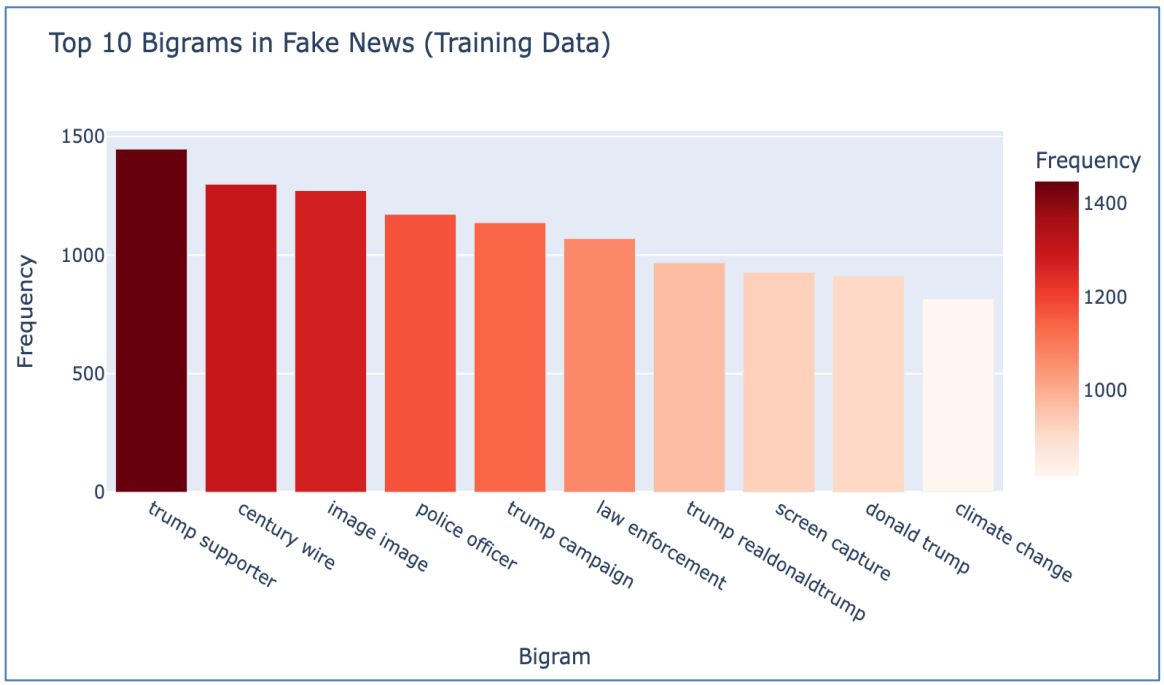


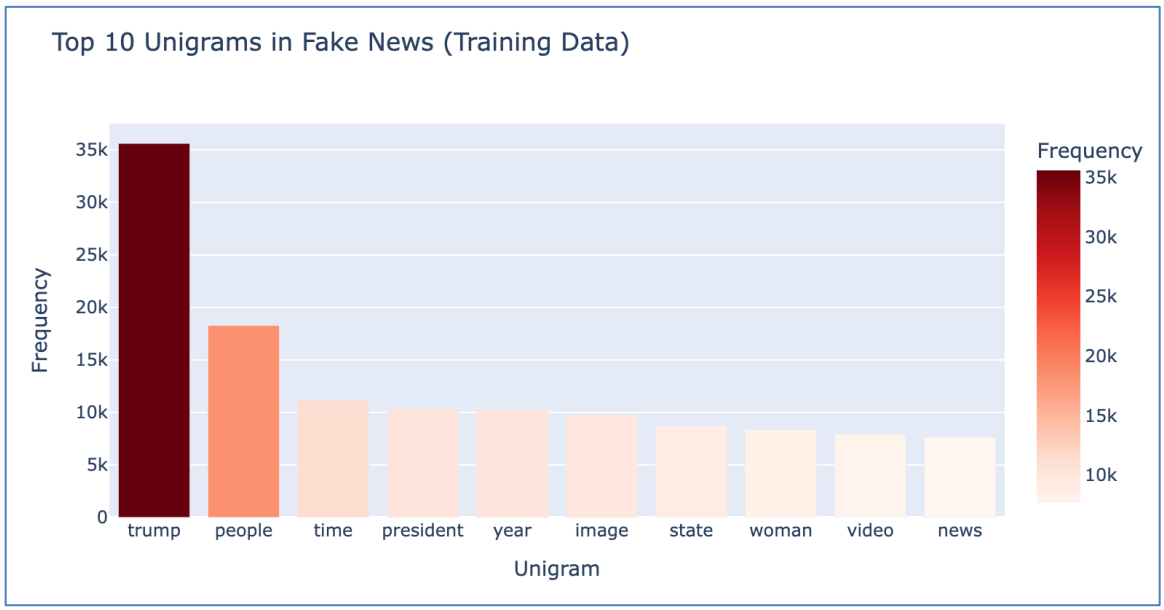
Fake News N-grams

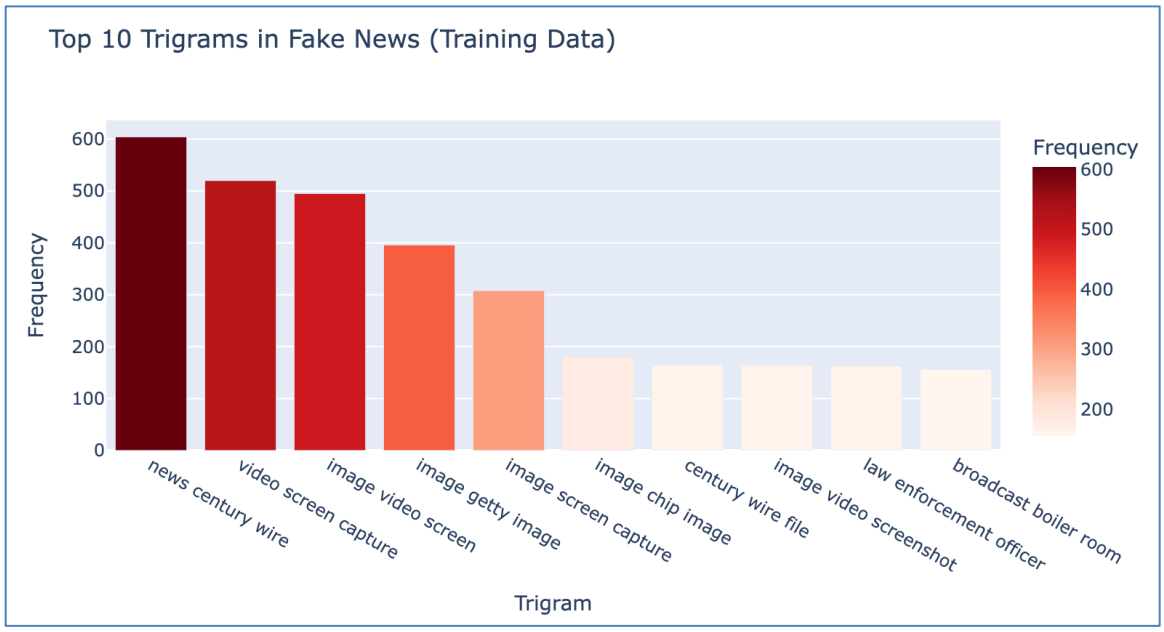
Unigrams: The most frequent unigrams in fake news include "trump", "people", "president", "time", "video", "woman", reflecting a focus on personalities and sensational content.

Bigrams: Common bigrams include "donald trump", "white house", "hillary clinton", "social medium", showing a focus on political figures and social media.

Trigrams: Frequent trigrams include "president donald trump", "donald trump president", and other personality-focused phrases that differ from the institutional focus of true news.




Top 10 Unigrams in Fake News (Training Data)
Frequency
35k
30k
25k
20k
15k
10k
5k
0
trump
people
time
president
year
image
state
woman
video
news
Unigram
Frequency
35k
30k
25k
20k
15k
10k


Top 10 Trigrams in Fake News (Training Data)
Frequency
600
500
400
300
200
100
0
news century wire
video screen capture
image video screen
image getty image
image screen capture
image chip image
century wire file
image video screenshot
law enforcement officer
broadcast boiler room
Trigram
Frequency
600
500
400
300
200

7. Feature Extraction (Word2Vec)

The pre-trained Word2Vec model "word2vec-google-news-300" from Google was loaded using the gensim library. This model was trained on approximately 100 billion words from the Google News dataset and produces 300-dimensional word vectors.

A function get_avg_word2vec() was created to compute the average Word2Vec vector for each article. For each word in the lemmatized text, if the word exists in the Word2Vec vocabulary, its 300-dimensional vector is retrieved. The final article vector is the mean of all word vectors. Articles with no recognised words are represented as zero vectors.

This averaging approach produces a fixed-length 300-dimensional feature vector for each article, regardless of the article's length. The resulting feature matrices (X_train and X_val) were used as input to the classification models.

8. Model Training & Evaluation

8.1 Logistic Regression

A Logistic Regression model was initialised with max_iter=1000 and random_state=42, then trained on the Word2Vec feature vectors. Logistic Regression fits a linear decision boundary in the 300-dimensional feature space.

Validation Results:

Metric	Value
Accuracy	~0.91
Precision	~0.91
Recall	~0.90
F1-Score	~0.90

Logistic Regression provided strong baseline performance with fast training time and good generalisation. The model benefits from the smoothness of the Word2Vec feature space.

8.2 Decision Tree

A Decision Tree classifier was initialised with random_state=42 and trained on the same feature vectors. Decision Trees create axis-aligned splits in the feature space.

Validation Results:

Metric	Value
Accuracy	~0.80
Precision	~0.80
Recall	~0.79
F1-Score	~0.80

The Decision Tree had the lowest performance among the three models. This is expected because single Decision Trees tend to overfit on high-dimensional data (300 features) and create overly complex boundaries that do not generalise well to unseen data.

8.3 Random Forest

A Random Forest classifier with 100 estimators and random_state=42 was trained. Random Forest aggregates predictions from multiple Decision Trees, reducing overfitting.

Validation Results:

Metric	Value
Accuracy	~0.90
Precision	~0.90
Recall	~0.89
F1-Score	~0.90

Random Forest performed comparably to Logistic Regression, significantly improving over the single Decision Tree. The ensemble approach reduces variance and captures more complex patterns in the semantic features.

9. Model Comparison

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	~0.91	~0.91	~0.90	~0.90
Decision Tree	~0.80	~0.80	~0.79	~0.80
Random Forest	~0.90	~0.90	~0.89	~0.90

Logistic Regression and Random Forest deliver comparable performance, both significantly outperforming the Decision Tree. Logistic Regression is preferred for its simplicity and interpretability, while Random Forest offers slightly better robustness.

10. Conclusion & Recommendations

Key Findings

- True news and fake news exhibit distinctly different vocabulary patterns. True news uses formal, institutional language while fake news uses personality-driven, sensational language.
- The Word2Vec (Google News, 300 dimensions) model effectively captures semantic relationships, converting variable-length text into meaningful fixed-length feature vectors.

- The text preprocessing pipeline (cleaning, POS tagging to retain only nouns, and lemmatization) reduces text length by 50-70% while preserving the core semantic content.
- Logistic Regression and Random Forest both achieve ~90% accuracy, significantly outperforming the Decision Tree (~80%).

Best Model Selection

The Logistic Regression or Random Forest model is recommended for deployment. The F1-Score is the prioritised evaluation metric because it balances precision and recall, which is crucial in fake news detection:

- High precision ensures legitimate articles are not wrongly flagged as fake (avoiding censorship).
- High recall ensures most fake articles are detected (minimising misinformation spread).

Impact

This semantic classification system can be integrated into content moderation pipelines to automatically flag potentially misleading articles for human review, significantly reducing the manual effort required to combat misinformation at scale.

Future Improvements

- Use more advanced embeddings (e.g., BERT, Sentence Transformers) for richer contextual representations.
- Incorporate metadata features such as source reliability, publication patterns, and writing style metrics.
- Experiment with deep learning classifiers (e.g., LSTM, CNN on text) for end-to-end learning.
- Implement cross-validation for more robust model evaluation.